Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
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(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

ScanPoint: A Real-Time ID Detection system using ML

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Project Guide
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Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement and Objective
- Proposed System
- Experimental Setup(Details about input/ dataset, Performance evaluation parameters, Software and hardware setup)
- Implementation(Screenshots of all modules)
- Conclusion and future work
- References
- Research Paper Publication / Patent/ Copyright status

Introduction

- ID verification is critical in banking, traffic policing, HR, rentals, etc.
- Manual verification is slow, inconsistent & prone to error.
- Rise of fake IDs increases fraud and security risks.
- Current solutions:
  - Standalone OCR tools :- text only no validation
  - Manual cross-checks :- time-consuming
  - Proprietary KYC systems :- Expensive & sector-specific
- Gap: Few systems combine OCR + database validation + AI face matching in one pipeline.

Introduction

- Observed delays at checkpoints, banks, HR offices.
- Difficulty verifying if the person matches the photo on the ID.
- Poor image quality makes manual reading harder.
- Need for quick, accurate and scalable verification.
- Automated OCR extraction of ID details.
- Regex-based data parsing and normalization.
- Database validation for legitimacy (“legit / fake / unknown”).
- AI-powered face detection and matching.
- Quality checks for low-resolution or blurred images.
- Clean Streamlit UI for non-technical users

Literature Survey of the existing system

Name of the research paper	Authors	Published on	Summary	Idea Adopted
Automated ID Card Detection and Penalty System Using YOLOv5 and Face Recognition	Anonymous (ResearchGate)	Jun 12, 2024	Proposes a real-time system that detects whether an ID card is worn using YOLOv5, and if not, applies face recognition to identify the person.	Inspired us to combine **object detection + face recognition** for enforcing ID compliance.
Real-time Facial Recognition via Multitask Learning on Raspberry Pi	Aboluhom et al.	Feb 16, 2025	Introduces a hybrid CNN-Transformer with structural reparameterization for fast, accurate face recognition on mobile devices	Motivated our idea to use multitask **learning** to improve recognition efficiency on low-power systems
FaceLiVT: Lightweight Vision Transformer for Real-Time Face Recognition	Setyawan et al.	Nov 17, 2022	Uses spectral imaging instead of RGB to improve live face detection under challenging lighting conditions	Encouraged us to consider alternative imaging modalities for robust detection in poor environments
Accurate Real-Time Live Face Detection Using Snapshot Spectral Imaging	Research Group (RG)	2025		

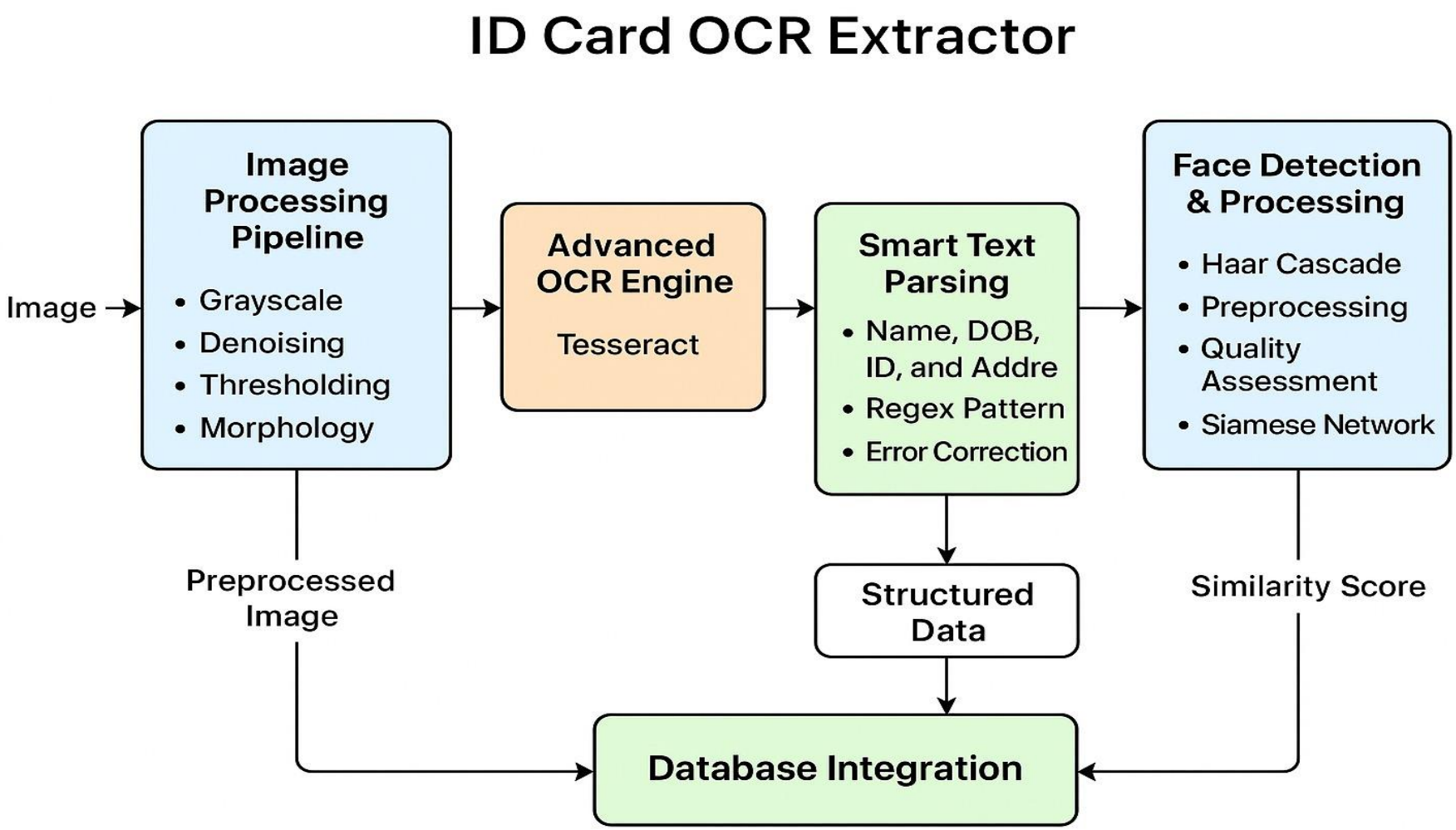
Limitations of Existing Systems

Existing Solutions	Issues / Limitations
Manual ID verification at counters & checkpoints	Slow, inconsistent, prone to human error
Standalone OCR apps (text only, no validation)	Low OCR accuracy on poor-quality images; fragmented
Bank/Insurance KYC apps with limited scope	Limited to specific sectors; not customizable
Third-party ID verification services (Onfido, Jumio)	High cost, privacy concerns, mostly cloud-based
Government Digital Identity Platforms (Aadhaar, DigiLocker)	Dependence on central infrastructure; no face matching

Problem statement

- Manual ID verification is slow, error-prone & inconsistent.
- Existing apps either do OCR or face matching not both.
- High fraud risk & poor user experience in current systems.
- Proposed: integrated OCR + face matching + database validation for fast, secure ID verification.

System Design



Technologies and methodologies

- OpenCV Preprocessing
  - Grayscale, denoising, Otsu & adaptive thresholding, morphological ops.
  - Our customization: Multi-variant pipeline to handle poor quality IDs, auto-selects best result.
- Tesseract OCR
  - OCR engine with configurable PSM/OEM modes; regex-based parsing.
  - Our customization: Automatic error correction (O→0, I→1), Indian state code mapping, auto uppercasing.
- Regex Parsing
  - Extracts DOB, vehicle/ID numbers, multi-line addresses.
  - Our customization: Support for multiple Indian formats + heuristics for name detection.

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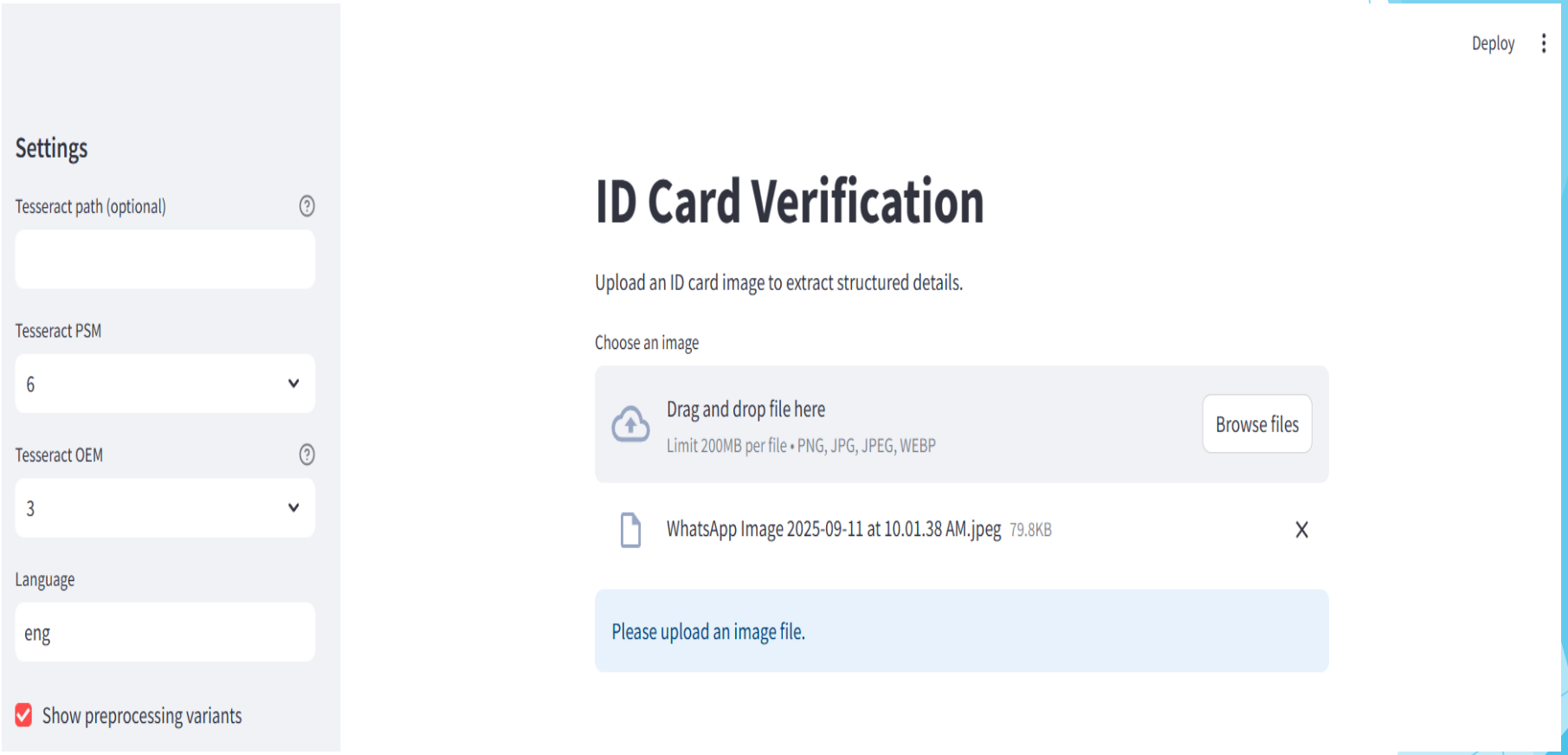
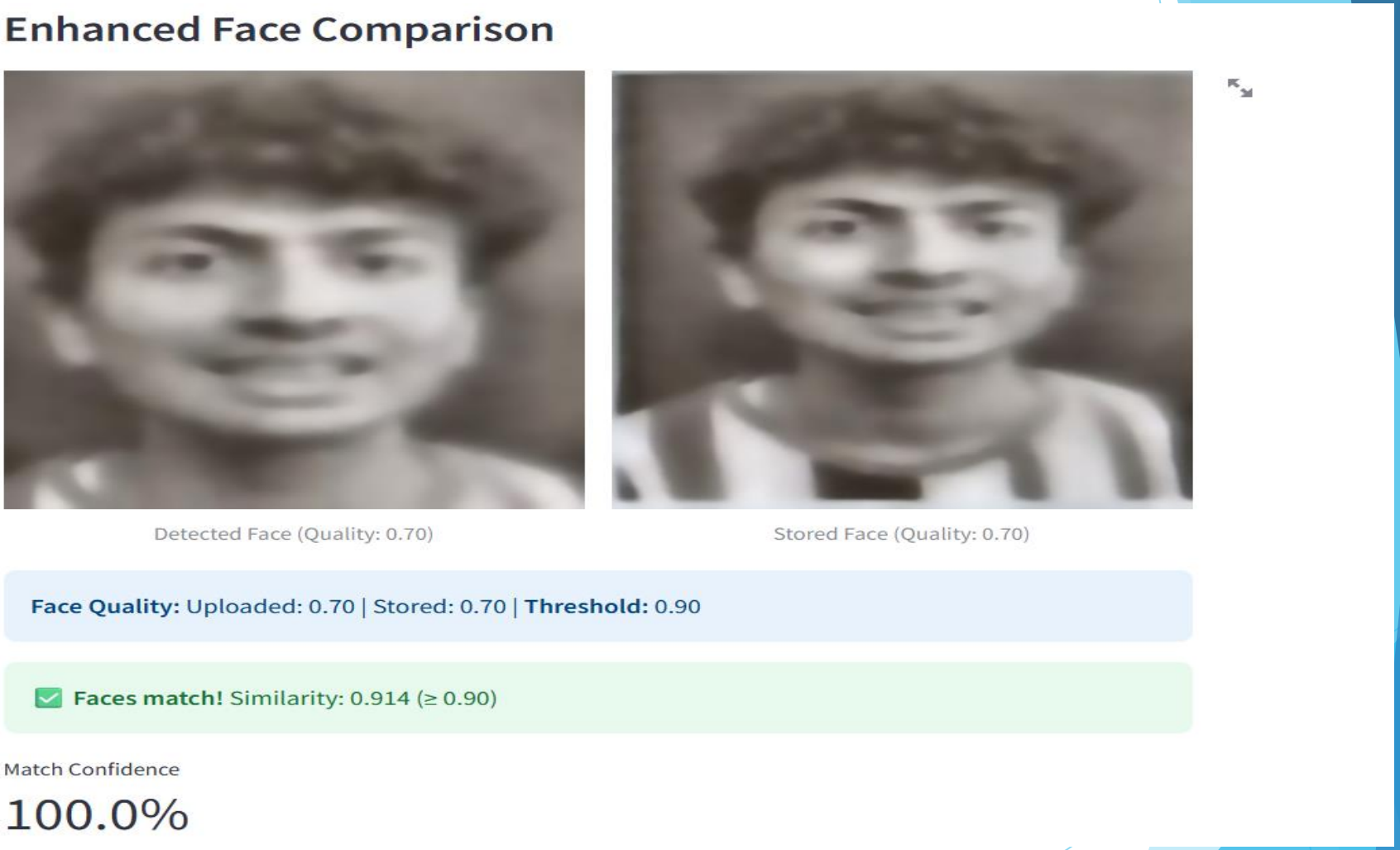
Technologies and methodologies

- Database Validation (PostgreSQL / Supabase)
  - Verifies extracted IDs against a preloaded database.
  - Our customization: Hardcoded Indian DL codes; returns legit/fake/unknown; fails gracefully on connection errors.
- Face Detection & Matching
  - Haar Cascade + Siamese Network (ResNet backbone, L2 normalization).
  - Our customization: Adaptive thresholds based on quality; multi-scale testing; quality scoring & rejection for low-quality faces.
- Streamlit UI
  - Real-time processing, side-by-side previews, JSON export.
  - Our customization: Minimalist, professional interface with live debugging of preprocessing.

Implementation



Expected Outcome



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Thank You...!!